

Handling Categorical vs Numerical Data Questions

Conceptual Questions

1. Explain the fundamental differences between categorical and numerical data. Provide three examples of each type and explain why they belong to that category.
2. Compare nominal and ordinal categorical data. How does the analysis approach differ between these two types? Provide specific examples of analytical techniques for each.
3. What is cardinality in categorical data and why is high cardinality problematic? Describe three strategies to handle high cardinality features.
4. Differentiate between discrete and continuous numerical data. How do the statistical measures and visualization techniques vary between them?
5. Explain the concept of data distribution in numerical variables. What are the key characteristics you examine when analyzing a distribution?

Technical Implementation Questions

6. Given a dataset with mixed data types, write Python code to automatically separate categorical and numerical columns and generate appropriate summary statistics for each.
7. How would you detect and handle inconsistent formatting in categorical data (e.g., "Male", "M", "male" all representing the same category)? Provide code examples.
8. Describe three different methods to identify outliers in numerical data and implement one method using Python.
9. You have a categorical variable with 15 unique categories. What visualization techniques would you use to analyze it and why? Provide code for at least two different plots.

10. For a numerical variable with significant skewness, what transformation techniques would you consider? Explain when you would use each transformation.

Analysis & Interpretation Questions

11. You're analyzing a "Customer Rating" variable (scale 1-5) and an "Age" variable. Which statistical measures are most appropriate for each and why?

12. How would you determine if a numerical variable follows a normal distribution? Describe both visual and statistical methods.

13. When analyzing categorical data, when would you use frequency tables vs. proportion tables? Provide specific scenarios for each.

14. You discover that a numerical variable has a bimodal distribution. What does this suggest about your data and what additional investigations would you perform?

15. Compare the use of histograms, box plots, and violin plots for numerical data analysis. When is each most appropriate?

Scenario-Based Questions

16. You're analyzing survey data with Likert scale questions (Strongly Disagree to Strongly Agree). How would you handle this ordinal data in your analysis?

17. A dataset contains ZIP code information as categorical data with 500 unique values. How would you handle this high-cardinality feature in your EDA?

18. You have a numerical variable "Income" that ranges from \$20,000 to \$2,000,000 with most values clustered around \$50,000. How would you analyze and visualize this heavily skewed data?

19. Working with medical data, you have both categorical (diagnosis codes) and numerical (lab results) variables. How would you analyze relationships between these different data types?

20. Your dataset has a "Timestamp" column. How would you extract meaningful categorical and numerical features from this datetime information?
